PROJECT AND TEAM INFORMATION

Project Title

Disaster Response and Resource Management System

Student / Team Information

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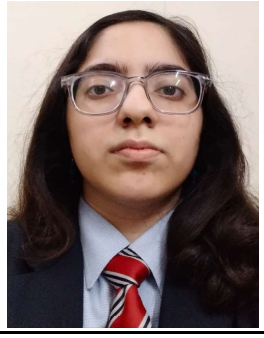
Ria Kataria (Team Lead)	*Name: Ria Kataria* *Student id: 23022107* *Email: 23022107@geu.ac.in* 
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PROPOSAL DESCRIPTION

Motivation

The motivation of this project is to help coordinate people in times of disaster where people may have trouble finding shelters, hospitals, and help. This project intends to provide one single place for reporting emergencies, to enable volunteers and NGOs to respond to and manage disasters quickly.

State of the Art / Current solution

Disaster management is currently done using government portals and helpline number. There is no common platform for coordination of users, volunteers and NGOs during disasters. There is a need for a role-based platform for better resource management in times of disaster.

Project Goals and Milestones

1. Design and implement a centralized Disaster Management System. It will have role based access for Admin, Volunteer, NGO and User.

2. Implement map-based resource tracking and emergency reporting features.

3. Use frontend (React.js), backend (Node.js), and database (PostgreSQL).

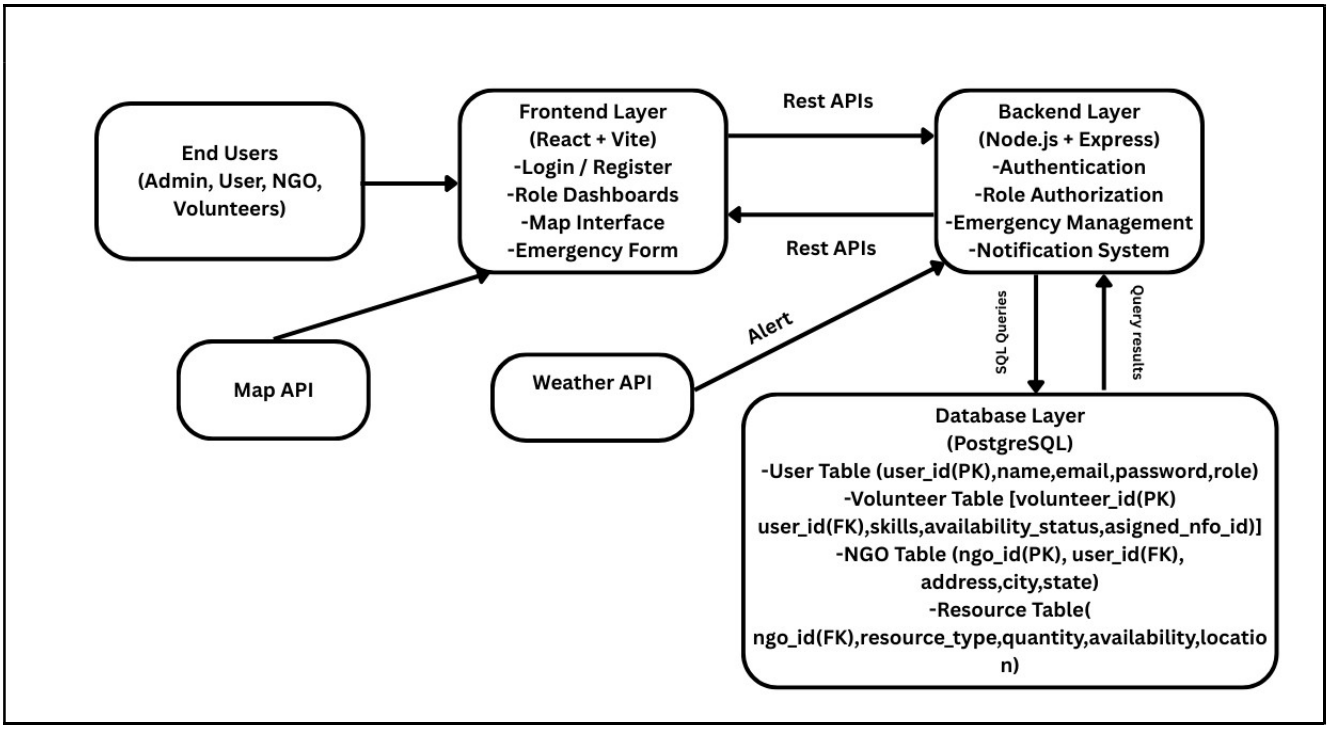
Project Approach

1. Implement role based and different dashboards for each role.

2. Integrate map services into the backend and database.

3. Implement features for users to report emergencies back to the backend and database.

System Architecture



Project Outcome / Deliverables

1. Disaster Management System with dashboards based on different roles.

2. Map-based resource tracking and emergency notifications.

3. Backend APIs integrated with PostgreSQL database.

Assumptions

This project is based on the assumption that all users have access to internet and computer, so that they use the website during an emergency. It is also assumed that users have location services turned on in order to give accurate map locations and help rescue teams locate them, also when users report an emergency they will give accurate information.

References

PostgreSQL Documentation (postgresql.org)- *https://www.postgresql.org/docs/*

React + Vite (vite.dev) - *https://vite.dev/guide/*

National Disaster Management Authority official website- *https://ndma.gov.in/*